SACHIN ABEYWICKRAMA

SOFTWARE ENGINEERING UNDERGRADUATE

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Software Engineering undergraduate with hands-on experience building web, Java, Python, and AI-integrated applications. Strong foundation in object-oriented programming, databases, algorithms, full-stack development, and practical problem-solving. Seeking developer roles where I can build reliable, user-focused systems and contribute effectively in collaborative engineering teams.

AREAS OF EXPERTISE

- Full-Stack Web Development
- Object-Oriented Programming
- React Application Development
- AI/ML Integrated Systems
- Frontend Development
- Database Design
- Problem-Solving
- UI/UX Prototyping
- Backend/API Development
- Java Application Development
- Git & Version Control
- Team Collaboration

PROFESSIONAL EXPERIENCE

SDB Bank **June 2024 - June 2025**

Software Engineering Intern

- Contributed to the design, development, and maintenance of internal enterprise applications, including the Asset Tracking System and Bank Intranet.
- Worked with Spring Boot for backend APIs, Angular/TypeScript and SCSS for frontend development, and MySQL for application data management.
- Developed frontend UI components, integrated RESTful APIs, and supported secure authentication and role-based access features.
- Used Git/GitHub for version control and collaborated with senior developers across requirement gathering, UI design, debugging, testing, and deployment activities.
- Supported software deployments by performing post-deployment checks, collecting user feedback, resolving issues, and preparing simple user guidance where required.

EDUCATION

University of Westminster **September 2022 - May 2026**

Bachelor of Engineering (Honours) in Software Engineering

Thesis: An IoT-Enabled Machine Learning-Based Water Quality Monitoring, Forecasting, and Decision Support System for Aquaculture

Supervisor: Imesh Pathirana

Key Modules: Object Oriented Programming, Concurrent Programming, Database Systems, Algorithms, Server-side Web Development, Operational Research Methods, Web Designing, Cybersecurity

Memberships: IEEE Club, Rotaract Club, IEEE Computer Society

Royal College, Monaragala **2012 - 2021**

- Passed GCE Advanced Levels in 2021 in the Mathematics Stream (Mathematics, Physics and ICT)
- Passed GCE Ordinary Levels in 2018

Memberships: President of the Literary Society, Senior Prefect of Royal College Monaragala, Member of the Interact Club, and Member of the Badminton Team

KEY PROJECTS

AquaAdapt: Water Quality Prediction System for Aquaculture Farmers (Python, FastAPI, Next.js, React, PyTorch, TypeScript, Tailwind CSS, Firebase, Open-Meteo API)
Final year research project focused on developing a full-stack aquaculture water-quality monitoring and forecasting system. I developed a web dashboard that predicts future pH, temperature, and turbidity using deep learning models, integrates weather data, and provides alerts, recommendations, model validation, and explainable AI support for better pond management.

GestureSway: Hand gesture-based snake game (Python, React, MongoDB, Node.js)
I've developed a web-based hand-gesture-controlled snake game using machine learning and the MERN stack. I've designed to support users with hand mobility limitations through interactive gesture-based gameplay

Thread-Safe Hospital Management System (Java, Concurrent Programming)
I've built a concurrent hospital management system using Java to simulate and manage multiple hospital operations safely and efficiently, with a focus on synchronisation, thread safety, and reliable handling of shared resources.

Westminster Shopping Manager (Java, Java Swing)
An online shopping system enabling managers to add products for sale and providing users with a Java Swing-A graphical interface for browsing and purchasing items added by managers.

TECHNICAL AND PROGRAMMING PROFICIENCIES

Programming Languages
Python, Java, JavaScript, TypeScript

Frameworks
React, Next.js

Machine Learning
PyTorch, Deep Learning, Time-Series Forecasting, Data Preprocessing, Model Training, Model Evaluation, Backtesting, Explainable AI, Ensemble Models

Cybersecurity
Kali Linux, OWASP fundamentals, Network security concepts